

**AMENDMENT NO. 3 MARCH 2026**

**TO**

**IS 1659 : 2004 — BLOCK BOARDS — SPECIFICATION**

*(Fourth Revision)*

*(Page 1, clause 4.1)* — Substitute the following for the existing:

‘**4.1** Block boards shall be of the following three grades:

- a) MR (moisture resistant) Grade;
- b) BWR (boiling water resistant) Grade; and
- c) BWP (boiling water proof) Grade.’

*(Page 1, clause 4.3)* — Substitute the existing grades and types of block boards with the following:

| <i>Grade and Type</i>      | <i>Symbol</i> |
|----------------------------|---------------|
| MR Grade, Decorative type  | MR-DEC        |
| MR Grade, Commercial type  | MR-COM        |
| BWR Grade, Decorative type | BWR-DEC       |
| BWR Grade, Commercial type | BWR-COM       |
| BWP Grade, Decorative type | BWP-DEC       |
| BWP Grade, Commercial type | BWP-COM       |

*(Page 1, clause 4.3)* — Insert the following new clauses **4.4** and **4.5** after **4.3**:

‘**4.4** In terms of bending class, each grade and type of block board shall be further classified in bending classes as follows:

- a) Bending Class 25
- b) Bending Class 40
- c) Bending Class 50’

‘**4.5** In terms of formaldehyde class, each grade and type of block board shall be further classified in bending classes as follows:

- a) Formaldehyde Class  $E_1$ ; and
- b) Formaldehyde Class  $E_2$ .’

*(Page 1, clause 5.1.1)* — Substitute the following for the existing:

‘**5.1.1** Any suitable species of timber and species of wood from other sources than natural forests such as wood from rubber, cashew, industrial and social forestry plantations, and shade trees from tea and coffee estates, may be used for the manufacture of block boards. The list of species given in Annex B is for guidance only.’

*(Page 1, clause 5.1.2, para 2)* — Substitute the following for the existing.

‘Species of wood from sources other than natural forests such as wood from rubber, cashew industrial and social forestry plantations, etc, and shade trees from tea and coffee estates may also be used.’

**Price Group 3**

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(Page 1, clause 5.2) — Substitute the following for the existing:

‘5.2 For MR Grade block boards, the adhesives shall be MR type conforming to IS 848. For BWR Grade block boards, the adhesives shall be BWR type conforming to IS 848. The adhesives used for bonding purposes shall be the BWP type conforming to IS 848 for BWP Grade block boards.’

(Page 2, clause 6.3, line 2) — Substitute ‘shall be sufficiently smooth’ for ‘shall be smooth’

(Page 2, clause 6.3.1) — ‘Delete’.

(Page 2, clause 6.3.3) — Substitute the following for the existing:

‘6.3.3 Permissible defects and tolerances on thickness of cross band and face veneers shall conform to IS 303 and IS 1328 for commercial and decorative veneers, respectively.’

(Page 2, clause 7.3) — Substitute the existing tolerances for length, width, and variation in thickness on a board with the following:

| <i>Dimension</i>                  | <i>Tolerance</i>     |
|-----------------------------------|----------------------|
| Length                            | +6 mm<br>–3 mm       |
| Width                             | +3 mm<br>–1 mm       |
| Variation in thickness on a board | Not more than 1.0 mm |

(Page 3, clause 9.2.2.2) — Insert the following new clause 9.2.2.3 after 9.2.2.2:

‘9.2.2.3 Test specimens from BWR grade block boards, shall be submerged in boiling water for 8 h and then tested as specified in Annex G, the test specimens shall comply with the requirements of 9.2.3.’

[Page 3, clause 9.2.4, (see also Amendment No. 1)] — Substitute the following for the existing:

‘9.2.4 Mycological Test.

Block board specimens, when tested according to Annex J shall show no visible signs of separation at the edges.’

(Page 3, clause 9.2.5) — Substitute the following for the existing:

‘9.2.5 Modulus of Rupture and Modulus of Elasticity.

The modulus of rupture and modulus of elasticity when tested according to the method given in Annex K, block boards shall have average values as given in Table 1. Also, no individual value shall be less than 90 percent of the average value specified for the respective bending class.’

**Table 1 Average Values of Modulus of Elasticity (*MoE*) and Modulus of Rupture (*MoR*)**

(Clause 9.2.5)

| SI No. | Properties                                                                | Bending Class |       |       |
|--------|---------------------------------------------------------------------------|---------------|-------|-------|
|        |                                                                           | 25            | 40    | 50    |
| (1)    | (2)                                                                       | (3)           | (4)   | (5)   |
| i)     | Modulus of Elasticity ( <i>MoE</i> ), N/mm <sup>2</sup> , Avg, <i>Min</i> | 2 500         | 4 000 | 5 000 |
| ii)    | Modulus of Rupture ( <i>MoR</i> ), N/mm <sup>2</sup> , Avg, <i>Min</i>    | 25            | 40    | 50    |

NOTE — The bending class shall be expressed with respect to minimum average value average value to the nearest lower multiple of 10. However, when the minimum average value of Modulus of Rupture (*MoR*) or Scaled Modulus of Elasticity (*MoE*/100) is less than 30, the Bending Class may be expressed by flooring the value to the nearest lower multiple of 5.

Calculation:

$MoR \parallel$  = Modulus of Rupture along the grain (N/mm<sup>2</sup>)

$MoE \parallel$  = Modulus of Elasticity along the grain (N/mm<sup>2</sup>)

Scaled MoE values:

$$MoE \parallel * = MoE \parallel / 100$$

$V_{min}$  = Minimum value of variables

$$V_{min} = \text{Min} (MoR \parallel, MoE \parallel *)$$

The following expression defines the floored value of the Bending Class (BC) based on the minimum value  $V_{min}$ :

- Flooring to 10s (default),  $BC_{10} = 10 \times \lfloor V_{min} / 10 \rfloor$
- Flooring to 5s (when  $V_{min} < 30$ ),  $BC_5 = 5 \times \lfloor V_{min} / 5 \rfloor$

where,

$V_{min}$  = minimum value of the variable under consideration.

$\lfloor \rfloor$  = the floor function, which rounds down to the nearest integer.

Example calculation for determining Bending class is as given below:

Example 1;

$$MoR \parallel = 38.3 \text{ N/mm}^2$$

$$MoE \parallel = 4200 \text{ N/mm}^2 \rightarrow MoE \parallel * = 42 \text{ N/mm}^2,$$

$$V_{min} = \text{Min} (38.3, 42) = 38.3$$

$$\text{Bending Class using flooring to 10s, } BC_{10} = 10 \times \lfloor 38.3/10 \rfloor = 10 \times 3 = 30$$

Example 2;

$$MoR \parallel = 28.3 \text{ N/mm}^2$$

$$MoE \parallel = 4100 \text{ N/mm}^2 \rightarrow MoE \parallel * = 41 \text{ N/mm}^2$$

$$V_{min} = \text{Min} (28.3, 41) = 28.3 \text{ N/mm}^2$$

$$\text{Bending Class using flooring to 5s, } BC_5 = 5 \times \lfloor 28.3/5 \rfloor = 5 \times 5 = 25.'$$

[Page 3, clause **9.2.6**, line 1 (see also Amendment No. 1 and 2)] — Insert 'as given in **6.1**' after 'preservative treatment'

[Page 3, clause **9.2.6**, (see also Amendment No. 1 and 2)] — Insert the following new clauses **9.2.7** and **9.2.8** after **9.2.6**:

**'9.2.7 Formaldehyde Content Test**

Test of free formaldehyde content in the sample shall be done by perforator method as per IS 13745 and test results of drawn sample shall meet the requirements of formaldehyde content as follows:

- Formaldehyde Class,  $E_1 < 8 \text{ mg/ 100 g}$  of oven dry sample; and
- Formaldehyde Class,  $E_2 \leq 30 \text{ mg/100 g}$  of oven dry sample.'

**'9.2.8 Steady-State Formaldehyde Emission Test (Optional Test).**

Manufacturer will offer a stack of block boards made under a batch (not less than 50 pieces) bearing the batch number and date of manufacture along with details of block board, that is, grade, size and thickness, adhesive used and any other (the manufacture wish to inform). After sampling, the sample shall immediately cover in airtight polythene with date of sampling on the block board.

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Sample must be taken for test within 24 h of opening the airtight cover and within one month from the date of sampling, and test shall be done by the test method as given in IS/ISO 12460 (Part 1). The test results of drawn sample shall meet the requirements of formaldehyde emission value as follows:

- a) Formaldehyde Class,  $E_1 < 0.124 \text{ mg/m}^3$ ; and
- b) Formaldehyde Class,  $E_2 > 0.124 \text{ mg/m}^3$ .

(Page 4, clause 11) — Substitute the following for existing:

#### **‘11 MARKING**

Each block board shall be legibly and indelibly marked or stamped with the following:

- a) Manufacturer’s name and his initials or his recognized trademark, if any
- b) In words ‘BLOCK BOARD’
- c) The grade and type (*see 4.3*)
- d) Bending class (*see 4.4*) for example, “Bending Class 25”
- e) Formaldehyde class (*see 4.5*)
- f) Species of timber
- g) Nominal length, width and thickness of block board
- h) Month and year of manufacture and
- j) Batch No.’

(Page 4, clause 11.1 and 11.1.1) —Substitute the following for existing:

#### **12 BIS Certification Marking**

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed there under, and the products may be marked with the Standard Mark.

(Page 4, Annex A) – Substitute the following for the existing:

| <i>IS No.</i>  | <i>Title</i>                                                                                                                     |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| IS 303 : 2024  | Plywood for general purposes — Specification ( <i>fourth revision</i> )                                                          |
| IS 401 : 2001  | Preservation of Timber — Code of Practice ( <i>fourth revision</i> )                                                             |
| IS 707 : 2011  | Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms ( <i>third revision</i> )                         |
| IS 848 : 2025  | Synthetic Resin Adhesives for Plywood (Phenolic, Aminoplastic and Bio-Materials Based) — Specification ( <i>third revision</i> ) |
| IS 1141 : 1993 | Seasoning of Timber - Code of practice ( <i>second revision</i> )                                                                |
| IS 1328 : 2025 | Veneered Decorative Plywood — Specification ( <i>fourth revision</i> )                                                           |
| IS 7638 : 1999 | Wood/Lignocellulosic based panel products—Methods of sampling ( <i>second revision</i> )                                         |

[Page 9, Annex G, references, (see also Amendment No. 1)] – Substitute the existing clause references under the Annex heading by the following:

‘(Clauses 9.2.2.1, 9.2.2.2, and 9.2.2.3)’.

[Page 9, Annex G, clause **G-1.2**, (see also Amendment No. 1)] — Insert the following new clause **G-1.3** after **G-1.2**:

‘**G-1.3** For testing of BWR grade block boards, the test specimens shall be submerged in boiling water for 8 h. The sample shall thereafter be tested as per **9.2.3**.’